

DHWANILNATH GAUTAM

(formerly Dhwaniilnath Gharekhan)

Ahmedabad, Gujarat, India | www.linkedin.com/in/dhwaniilnath |

dhwaniilnath@gmail.com | +91 89808 88847 |

www.cept.ac.in/faculty/dhwaniilnath-gharekhan | ORCID: 0000-0002-6699-7494

Research Manuscripts: 26 [1 Accepted, 10 published, 15 conference papers]



PROFESSIONAL EXPERIENCE

Orbrick Consulting Pvt. Ltd., Ahmedabad, India | Corporate Consultant

Mar 2024-Present

- Providing solutions for deploying Machine Learning in employment management systems for higher education institutions.

CEPT University, Ahmedabad, India | Program Chair & Assistant Professor

Aug 2022-Present

- Presiding Chair for the M. Tech. Geomatics program, focusing on curriculum development, outreach, and research and academic collaborations with other organizations.
- Teaching subjects including Geospatial Programming, GIS for Smart Cities, Applications of Spatial Big Data and Analytics, Geospatial Modelling and Applications, and Python for Statistics.
- Managing other institutional responsibilities for Faculty of Technology, including Admissions Committee Coordinator and Computer Lab Coordinator (elaborated below).

Nirma University, Ahmedabad, India | Lecturer

Aug 2018-Sep 2022

- Teaching subjects including Surveying, GIS, and Remote Sensing to students of B.E. (Civil Engineering) and providing guidance to students for minor projects and final thesis.

ISRO, Ahmedabad, India | Graduate Student Researcher

Jun 2018-Jan 2021

- Conducting primary research and developing computational models in the fields of Applied Meteorology, Thermal and Multispectral Remote Sensing, and Machine Learning at the Space Applications Centre (SAC), Indian Space Research Organisation (ISRO), Ahmedabad.

Gujarat University, Ahmedabad, India | Visiting Faculty

Jul 2017-Jun 2018

- Teaching subjects including Spatial Analysis and Statistics, Natural Resource Management, and GIS Lab to students of M.Sc. (Geoinformatics).

St. Xavier's College, Ahmedabad, India | Researcher

Oct 2017-Jun 2018

- Conducting primary research under the project "Quantization of Aerosol Radiative Forcing over an Urban Environment" at the Department of Physics and Electronics at St. Xavier's College.

DISSERTATIONS GUIDED AT CEPT UNIVERSITY

Bachelor's Thesis

2023-2024

- Thesis Title: "Cooling the Concrete Jungle: Geospatial Insights into Urban Heat Islands and Urban Built Forms"

Master's Thesis

2022-2023

- Thesis Title: "Modelling glacial retreat using Machine learning Techniques"
<https://portfolio.cept.ac.in/2023/S/ft/drp/ge4050-drp000804-spring-2023>

INSTITUTIONAL RESPONSIBILITIES AT CEPT UNIVERSITY

Admissions Coordinator for Faculty of Technology

Nov 2023-Present

- Admissions coordinator for Faculty of Technology (for 1 UG and 4 PG programs) and core member of the Central Admissions Committee for managing, promoting and organizing admissions for the 2024-2025 cycle.

Computer Lab Coordinator for Faculty of Technology

Jan 2023-Present

- Maintaining and upgrading the computer lab at Faculty of Technology, as well as updating software licenses and acquiring new software and products for the systems.
- Coordinating with different faculty-level programs and ensuring systematic usage of the lab.

Convocation Coordinator-2023 & 2024

Jan 2023 and 2024

- Organizing and planning the faculty-level convocation for graduating students.

Social Media Coordinator for Geomatics

Nov 2022-Present

- Managing M. Tech. (Geomatics) program's social media profiles and coordinating with student volunteers for regular sharing of programmatic updates, events, and general information about geomatics, including collation and verification of data being shared.

LEADERSHIP ROLES AND GUEST LECTURES

Guest speaker | Gujarat Institute of Disaster Management, Gandhinagar, India

- Invited twice to give a special session on “**GIS & Remote Sensing: A Vital Tool for Environmental Impact Assessment**” at two training programs (4 to 6 Mar 2024 and 29 to 30 Jan 2024) organized by Gujarat Institute of Disaster Management (GIDM) in collaboration with Gujarat Environment Management Institute (GEMI) on ‘Navigating Environmental Impact Assessment and Clearance Processes’.

International Land Use Symposium | CEPT University, Ahmedabad, India

- Organized a three-day symposium on International Land Use (4 to 6 Oct 2023) and session chair and moderator for all Open Data and Machine Learning Tracks.

Google Earth Engine Workshop | CEPT University, Ahmedabad, India

- Organized a one-day workshop (18 Mar 2023) for school and college students on Google Earth Engine platform at CEPT University.

GIS Day 2022 | CEPT University, Ahmedabad, India

- Organized a one-day workshop (15 Oct 2022) for school and college students on the theme of ‘Story maps’ as a part of the International GIS Day celebration hosted by ESRI.

EDUCATION

- Nirma University, Ahmedabad, India | Ph.D. in Geomatics** **2018-2022**
- CEPT University, Ahmedabad, India | M.Tech. in Geomatics** **2015-2017**
- Silver Oak University, Ahmedabad, India | B.E. in Electronics and Communication** **2010-2015**

SKILLS

Python, ENVI, MATLAB, ERDAS, QGIS, SBDART, ESRI Suite

ACHIEVEMENTS

Best Paper Presentation | National Symposium at ISRO, Ahmedabad

Dec 2018

- Third Best Presentation at the National Symposium on ‘Advancements in Geospatial Technology for Societal Benefits 2018’ organized by the Indian Society of Geomatics and the Indian Society of Remote Sensing at SAC, ISRO, Ahmedabad.

Distinguished Student 2018 | CEPT University, Ahmedabad

Jan 2018

- Honored with the award of "Distinguished Student" from M.Tech. Geomatics, batch of 2015-2017 at CEPT University.

First Prize–International Summer School | Wuhan University, China

Jun 2016

- Received First Prize for the final project competition at the International Geoinformatics Summer School at Wuhan University, China.

VOLUNTEER WORK

Manuscript reviewer for the following journals:

Jan 2019-Present

- TGRS (Transactions of Geoscience and Remote Sensing)*
- JISRS (Journal of the Indian Society of Remote Sensing)*
- ASTESJ (Advances in Science, Technology and Engineering Systems Journal)*
- GSRL (Geoscience and Remote Sensing Letters)*
- RSE (Remote Sensing of Environment)*

- *ARES (Atmospheric Research)*
- *TJDE (International Journal of Digital Earth)*

IEEE GRSS Editorial Team

2022-Present

- Editorial team member for the Institute of Electrical and Electronics Engineers (IEEE) Geoscience and Remote Sensing Society (GRSS)—Gujarat Section newsletter.

Organizer and Panel Moderator

Dec 2021

- Organizing committee member and panel moderator for the India Geoscience and Remote Sensing Symposium (InGARSS) 2021 held virtually.

Teaching Assistant

Aug and Sep 2021

- Volunteer for the NVIDIA Deep Learning Institute (DLI) Workshops “Fundamentals of Accelerated Data Science” and “Fundamentals of Accelerated Deep Learning”.

Student Organizer

Oct 2015 and Sep 2016

- Organizer for a series of lectures and activities as part of GIS week celebrated at CEPT University.

SOCIETIES

IEEE-GRSS | *Individual Member*

2018-Present

Indian Society of Geomatics (ISG) | *Life Member*

2018-Present

Indian Society of Remote Sensing (ISRS) | *Life Member*

2018-Present

Indian Aerosol Science and Technology Association (IASTA) | *Life Member*

2018-Present

International Society of Photogrammetry and Remote Sensing (ISPRS) | *Individual Member*

2017-Present

PUBLISHED MANUSCRIPTS

1. Vemuri, S., **Gautam, D.**, Gandhi, S. (2024). Glacier retreat delineation using machine learning and deep learning: A case of a lower Himalayan region. J. Earth Syst. Sci. 133(82). <https://doi.org/10.1007/s12040-024-02285-4>, link: <https://rdcu.be/dE32A>
2. **Gharekhan, D.**, Nigam, R., Bhattacharya, B.K., Desai, D., Patel, P. R. (2022). Estimating regional-scale daytime net surface radiation in cloudless skies from GEO-LEO satellite observations using data fusion approach. J. Earth Syst. Sci. 131(73). <https://doi.org/10.1007/s12040-021-01806-9>, link: <https://rdcu.be/ciQrD>
3. **Gharekhan, D.**, Bhattacharya, B. K., Desai, D., Patel, P. R. (2022). Modelling of Downwelling Longwave Radiation over Multiple Agro-Climate Settings of India Under Foggy Sky Conditions – A Neural Network Approach, Remote Sensing Applications: Society and Environment, 25(1), 100684. <https://doi.org/10.1016/j.rsase.2021.100684>
4. **Gharekhan, D.**, Bhattacharya, B. K., Desai, D., & Patel, P. R. (2021). Neural network-based approach for estimation of downwelling longwave radiation flux under cloudy-sky conditions. Journal of Applied Remote Sensing, 15(2), 024515. <https://doi.org/10.1117/1.JRS.15.024515>
5. Patel, P. H., Patel, N. P., Patel, P. L., **Gharekhan, D.**, Gharekhan, A. (2017). Wavelet Based Analysis of Soil Nutrients of Surendranagar District, Gujarat, India, International Journal of Humanities, Arts, Medicine and Sciences, 5(5), 49-54
6. **Gharekhan, D.**, Patel, P. L., Gharekhan, A. (2017), “Disharmonic Identification of EEG data through Wavelet Transformation”, International Journal of Research in Modern Engineering and Emerging Technology 5(9), 1-10
7. **Gharekhan, D.**, Gandhi, S., (2017), Studying the Land Use Change for Understanding Old European Colonies in Present Era of India. Journal of Environmental Science, Computer Science and Engineering & Technology, 7(1), 60-69
8. Patel, P. H., Patel, N. P., Patel, P. L., **Gharekhan, D.**, Gharekhan, A., (2017). Wavelet Based Analysis of Soil Nutrients of Surendranagar District, Gujarat, India, BEST: International Journal of Humanities, Arts, Medicine and Science, 5(5), 49-54
9. Gharekhan, A., **Gharekhan, D.** (2015). Enforcement of Human Rights at National and International Level, A Challenge: Pakistan and India’s Fisher Women’s Perspectives, Research Expo International Multidisciplinary Research Journal, 5(2), 44-52

10. Gharekhan, A., **Gharekhan, D.** (2014), Importance of Integrated (Manual Labour Intrinsic & Mechanised) Solid Waste Management Systems, International Journal of Engineering Sciences & Research Technology 3(7), 221-225

RESEARCH PRESENTED

1. Hussain, S. A., Parande, U., Patel, P., Bhatt, M., **Gharekhan, D.**, & Gandhi, S. (2023). A Multi-Criteria Decision Analysis for Locating Optimal Food Zones, Rest Points and Local Markets in Rajkot: Enhancing Urban Liveability and Efficiency for the Gig Economic Sector. ILUS – International Land Use Symposium 2023, Urban Analytics for Transforming Cities and Regions: Tools, Methods and Applications, 4th-6th October 2023 at CEPT University in Ahmedabad, India. Poster Presentation, doi: 10.5281/zenodo.10060261
2. Singh, V., Patwa, P., Parmar, P., Bhatt, M., **Gharekhan, D.**, & Gandhi, S. (2023). Improving Stormwater Infrastructure using Geospatial Techniques in Rajkot Smart City. ILUS - International Land Use Symposium 2023, Urban Analytics for Transforming Cities and Regions: Tools, Methods and Applications, 4th-6th October 2023 at CEPT University in Ahmedabad, India. Poster Presentation, doi: 10.5281/zenodo.10060373
3. Srivastava, N., Kanmani, K. S., Bhatt, M., **Gharekhan, D.**, & Gandhi, S. (2023). Geo-Analytical Approach to diseases vulnerability and high-risk space in Rajkot. ILUS - International Land Use Symposium 2023, Urban Analytics for Transforming Cities and Regions: Tools, Methods and Applications, 4th-6th October 2023 at CEPT University in Ahmedabad, India. Poster Presentation, doi: 10.5281/zenodo.10060225
4. Pastagiya, K., Patra, A., R, T., Bhatt, M., **Gharekhan, D.**, & Gandhi, S. (2023). Suggestive improvements to the city's green cover for a sustainable future of Rajkot. ILUS - International Land Use Symposium 2023, Urban Analytics for Transforming Cities and Regions: Tools, Methods and Applications, 4th-6th October 2023 at CEPT University in Ahmedabad, India. Poster Presentation, doi: 10.5281/zenodo.10060466
5. S, A. P., Parackal, J. S., Bhatt, M., **Gharekhan, D.**, & Gandhi, S. (2023). Assessment of Rooftop Photovoltaic Potential For Rajkot Smart City. ILUS - International Land Use Symposium 2023, Urban Analytics for Transforming Cities and Regions: Tools, Methods and Applications, 4th-6th October 2023 at CEPT University in Ahmedabad, India. Poster Presentation, doi: 10.5281/zenodo.10030014
6. **Gharekhan, D.**, Bhattacharya, B. K. (2022). Prediction of land surface temperature under overcast skies using data fusion and deep learning neural networks, TRISHNADAYS, March 22-24, 2022, Toulouse & on-line everywhere, Oral Presentation
7. Chandelia, A., Patel, P. R., **Gharekhan, D.**, (2022). Hyperspectral Data Assimilation and Road Material Extraction, InGARSS 2021 Conference, 6-9 December, 2021, Ahmedabad, Gujarat, Oral Presentation
8. **Gharekhan, D.**, Bhattacharya, B. K., Nigam, R., Desai, D., & Patel, P. R. (2020). Modelling of daytime incoming longwave radiation under cloudy sky over a semi-arid agro-climatic zone of India. In Technologies for Sustainable Development (pp. 79-83). CRC Press. DOI: <https://doi.org/10.1201/9780429321573>, Oral Presentation, 7th Nirma University International Conference on Engineering, NUiCONE 2019, 21-22 Nov, 2019, Ahmedabad
9. **Gharekhan, D.**, Vyas, A., (2018, December), Built up construction material identification and classification from hyperspectral imagery. In National Symposium on Advancements in Geospatial Technology for Societal Benefits and Annual Conventions of Indian Society of
10. Geomatics & Indian Society of Remote Sensing 5-7 December, 2018, Ahmedabad, Gujarat., Oral Presentation
11. **Gharekhan, D.**, Turakhia, T., Iyer, R., Chabra, A., (2018) "Radiative forcing of special events over the urban environment of Ahmedabad City", Indian Aerosol Science and Technology Association (IASTA 2018) Conference @ IIT Delhi (26 to 28 Nov, 2018), Poster Presentation
12. **Gharekhan, D.**, Patel, P. L., Gharekhan, A. (2017), "Disharmonic Identification of EEG data through Wavelet Transformation", International Conference On "Recent Innovations In Science And Workshop Of Spectroscopy And Its Perspectives", Mehsana, Gujarat, Oral Presentation
13. **Gharekhan, D.**, Pramanik, N., Malik, A., Mughal, A., (2016), "Greece Crisis: A Case Study Between 2001 To 2015 Using Night Light Remote Sensing", International Geoinformatics Workshop Course "Data Mining of Night Light Remote Sensing and GNSS High Precision Navigation" Organised By Wuhan University, 19-30 June, 2016, Oral Presentation

14. Gharekhan, A., **Gharekhan, D.**, (2014), Enforcement of human rights at national and international level, A Challenge: Pakistan and India's fisher women's perspectives. International Conference on Contemporary Issues and Challenges of Human Rights in The Era of Globalization,
15. GLS, Ahmedabad, 2-3 August, 2014, Oral Presentation Gharekhan, A., **Gharekhan, D.**, (2014), Importance of Integrate (Manual Labor Intrinsic & Mechanized) Solid Waste Management Systems. National Conference on Integration of Engineering with Environmental Development & Sustainability; Pacific University, Udaipur, 15-16 March 2013, Oral Presentation